

QP Code : NP-18770

(3 Hours)

[Total Marks : 80]

N.B. : (1) Question No. 1 is compulsory.

(2) Solve any three questions out of remaining five.

1. (a) Define the following terms :-

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- (i) Transaction
- (ii) Primary key
- (iii) Deadlock
- (iv) Strong entity set
- (v) Lock point.

1. (b) Consider the following relation :-

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A	B	C	Tuple #
10	b ₁	c ₁	#1
10	b ₂	c ₂	#2
11	b ₄	c ₁	#3
12	b ₃	c ₄	#4
13	b ₁	c ₁	#5
14	b ₃	c ₄	#6

Given the previous state which of the following dependencies may hold in the above relation ? If the dependency cannot hold explain why by specifying the tuples that cause the violation :-

- (i) $A \rightarrow B$
- (ii) $B \rightarrow C$
- (iii) $C \rightarrow B$
- (iv) $B \rightarrow A$
- (v) $C \rightarrow A$

2. (a) Explain different data models with its advantages and disadvantages.

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(b) Explain Generalization, Specialization and Aggregation with the help of an example.

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3. (a) Construct an E-R diagram for a car-insurance company that has a set of customers each of whom owns one or more cars. Each car has associated with it zero to any number of recorded accidents.

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[TURN OVER]

(b) Define Deadlock Detection and Recovery.

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4. (a) Consider the following relations for a book club :-

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Members (Member-Id, Name, Designation, Age)

Books (Book-Id, Booktitle, BookAuthor, Bookpublisher, Bookprice)

Reserves (Member-Id, Book-Id, Date)

Write SQL queries for following statements :-

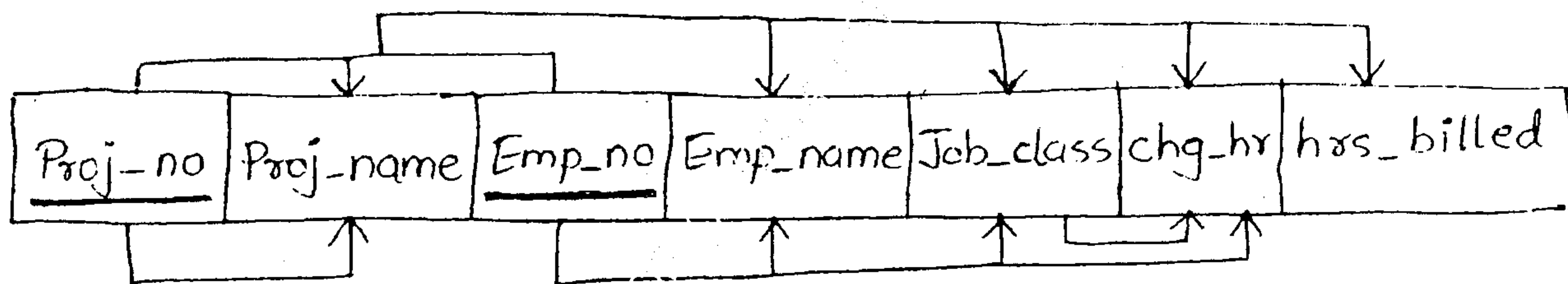
- (i) Find the names of members who are professor older than 50 years.
- (ii) List the titles of books reserved by professors.
- (iii) Find Ids of members who have reserved books that cost more than ₹ 500.
- (iv) Find the authors and titles of books reserved on 20-09-2012.

(b) What do you mean by serializability schedule ? How would you test whether given schedule S is conflict serializable.

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5. (a) Consider a dependency diagram of relation R and normalize it up to third normal form.

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(b) Explain shadow paging method.

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6. (a) Draw the Query tree for the following relational algebra expression :-

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π Customer - name (C branch - city = "Brooklyn" $\wedge$ balance > 1000

((branch $\bowtie$ (account $\bowtie$ depositor)))

(b) Explain the following relational algebra operations with proper examples :-

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- (i) Natural join
- (ii) Assignment
- (iii) Rename
- (iv) Set-Intersection operation
- (v) Union.